



Occupational Downgrading and Task Unbundling in High-Exposure AI Labor Markets

Predoc Training Series, Research Question Formulation

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1 Research Motivation

The conventional AI displacement literature evaluates labor market impact by tracking aggregate unemployment among highly exposed workers. That metric was calibrated for industrial-era technology shocks, and it systematically misses the primary adjustment mechanism in knowledge work. I hypothesize that generative AI displaces high-skill workers through three mutually reinforcing channels invisible to standard labor market statistics: *occupational downgrading*, where displaced professionals accept below-credential employment rather than registering as unemployed; *human capital pipeline depletion*, where AI codification of tacit knowledge eliminates the economic rationale for entry-level hiring and severs the career formation pathway for future graduates; and *contractual fissuring*, where AI task modularization converts salaried positions into contractor arrangements that appear as employment but lack the income continuity and insurance of formal employment. Together, these channels imply that the demand externality formalized by [Hemenway Falk and Tsoukalas \(2026\)](#) is already operating. However, it manifests through wage stagnation and balance-sheet erosion, rather than the unemployment spikes policy frameworks are calibrated to detect.

The AI labor market literature has converged on a tentatively reassuring reading of early data, showing that aggregate unemployment among highly AI-exposed workers has not risen significantly ([Massenkoff and McCrory, 2026](#)). While this observation is methodologically sound, using it to gauge the labor market security is misplaced because the diagnostic metric is misaligned with how knowledge workers adjust. The foundational task framework of [Autor et al. \(2003\)](#), extended by [Acemoglu and Restrepo \(2018, 2020\)](#), treats displacement fundamentally as a quantity problem driven by position-eliminating substitution. Generative AI is not that technology. As [Eloundou et al. \(2023\)](#) document, exposure is heavily concentrated among knowledge workers. These graduate-educated professionals possess the financial buffers and occupational flexibility that displaced manufacturing workers lacked. And when these workers face task substitution, they exercise that flexibility by downgrading into occupations that fall significantly below their credential and experience levels. None of these transitions register in the standard aggregate unemployment metrics, which are meant to capture outright job losses rather than structural wage and skill devaluation (e.g., [Massenkoff and McCrory, 2026](#)).

I term this the *measurement-welfare mismatch*, a condition where the standard indicator reads stable while the underlying welfare deterioration compounds. The stable unemployment figures in early AI data do not indicate that the trap has been averted. Instead, they demonstrate that the trap operates through a channel that existing monitoring infrastructure was not designed to capture. This research agenda aims to formalize the mechanisms driving this invisible channel and design the empirical strategy required to measure it before the economic damage becomes irreversible.

2 Research Question

This research agenda asks whether AI adoption in knowledge-intensive industries reduces occupational attainment among educated workers, and whether this downgrading channel accounts for the absence of an unemployment signal that [Massenkoff and McCrory \(2026\)](#) document despite measured declines in labor demand among high-exposure occupations. I propose addressing this question using a difference-in-differences design that links their Observed Exposure measure to longitudinal occupational trajectories in the Survey of Income and Program Participation (SIPP) and Current Population Survey (CPS). For identification, the empirical strategy exploits cross-industry variation in AI adoption timing, proxied by industry-level usage data from the Anthropic Economic Index ([Anthropic, 2025](#)). The proposed design specifically targets the three channels established in the motivation: occupational downgrading and credential-occupation mismatch in post-displacement employment, human capital pipeline depletion through the erosion of entry-level hiring, and contractual fissuring through the conversion of salaried positions into non-W-2 arrangements. Taken together, these channels are capable of generating income losses large enough to activate the demand trap formalized by [Hemenway Falk and Tsoukalas \(2026\)](#) while leaving aggregate unemployment entirely undisturbed. The *research question* therefore tests whether the apparent resilience of the knowledge-work labor market in the era of generative AI has increasingly diverged from traditional unemployment metrics, operating instead through a gradual human capital and income erosion that existing monitoring frameworks are structurally incapable of capturing.

3 Literature Review

3.1 Measurement Architecture and the Credential Mismatch Gap

The standard diagnostic of AI displacement inherits a structural misalignment from the task-based framework of [Autor et al. \(2003\)](#), extended by [Acemoglu and Restrepo \(2018, 2020\)](#) to formalize competing displacement and productivity effects. [Acemoglu and Restrepo \(2022\)](#) establish the macroeconomic backdrop by demonstrating that recent automation has been “so-so,” delivering substitution without sufficient new task creation to compensate. This dynamic accounts for most of the observed increase in US wage inequality over the past four decades. Both [Eloundou et al.’s \(2023\)](#) GPT substitutability scores and [Massenkoff and McCrory’s \(2026\)](#) Observed Exposure measure nevertheless treat labor market impact as fundamentally a *quantity* problem: has employment among highly exposed workers declined? This framing is appropriate for position-eliminating substitution, as with industrial robots, but wrong for a technology that operates through

task substitution *within* knowledge roles, where graduate-educated workers possess the credentials and social capital to absorb displacement through quality rather than quantity adjustment. [Autor \(2019\)](#) makes this point from the opposite direction, noting that “new work” created by technological change has heavily bifurcated between high-skill “frontier jobs” and low-skill “wealth work.” This structural hollowing-out raises competitive pressure on mid-level workers, forcing those displaced by automation to downgrade into lower-tier service roles rather than transitioning laterally.

The welfare cost of quality adjustment is documented in four decades of overeducation research. [Leuven and Oosterbeek \(2011\)](#) synthesize the overeducation literature and report that the cross-study central tendency for the credential-mismatch wage penalty is approximately 15%—a loss that accrues without any unemployment event. [Sicherman \(1991\)](#) documents that overeducated workers face a systematic wage discount relative to matched peers and, unless they exit through promotion or job-to-job mobility, remain in suppressed-wage positions over extended careers. Furthermore, [Pellizzari and Fichten \(2013\)](#), using PIAAC data across OECD economies, document that skill mismatch is widespread and cross-nationally durable, with little evidence that affected workers transition systematically back into matched employment. This represents a structural sorting failure rather than frictional adjustment, creating a stable, low-visibility trap that aggregate unemployment statistics cannot detect. Unifying these frameworks, [Autor and Handel \(2013\)](#) provide the direct bridge between the task and mismatch literatures. They demonstrate that occupational wages are governed by the task content of assigned roles rather than credential level alone, implying that credential-occupation mismatch suppresses earnings purely through organizational task allocation — without requiring any underlying deterioration of worker skill.

The gap in this literature is empirical rather than conceptual: no study links the intensity of AI adoption in a worker’s occupation to the credential-occupation match of their subsequent employment. Without this bridge, the [Massenkoff and McCrory \(2026\)](#) stability finding cannot be interpreted as evidence that displacement is absent, but rather as evidence that displacement may be operating entirely outside the boundaries of their measurement architecture.

3.2 Firm-Sponsored Training and Human Capital Pipeline Depletion

The more durable damage from generative AI is to the career formation infrastructure that produces the next generation of knowledge workers. The economic logic originates with [Becker \(1964\)](#)’s distinction between general and firm-specific human capital: firms invest in workers’ skills only when they can appropriate a share of the resulting surplus. [Acemoglu and Pischke \(1998\)](#) refine this framework by showing that the current

employer’s informational advantage over outside firms creates ex post monopsony power, generating private incentives to finance even general training; [Acemoglu and Pischke \(1999\)](#) extend the argument to show that wage compression under a variety of imperfect labor market conditions sustains firm-sponsored training as an equilibrium phenomenon across institutional contexts. [Topel \(1991\)](#) and [Neal \(1995\)](#) document that the resulting seniority premium is substantial and partly industry-specific, proving that this on-the-job accumulation of tacit capital is the primary engine of a worker’s lifetime earnings trajectory. Entry-level roles in knowledge-intensive organizations serve as the crucible for this formation, where skills like structuring client analysis or diagnosing software failures are acquired through learning-by-doing and no other route.

[Brynjolfsson et al. \(2023\)](#) demonstrate how generative AI disrupts this exact formation process. Studying 5,179 customer-support agents, they find a 14% average productivity gain that rises to 34% for novice workers because the AI codifies and disseminates the tacit practices of top performers. While typically interpreted as labor augmentation, this dynamic fundamentally eliminates the economic rationale for junior hiring. Once experienced-worker knowledge is embedded directly into the AI system, the joint surplus from the training relationship collapses alongside the entry-level pipeline.

The empirical literature documents the severe consequences of such entry-level hiring contractions. Studying recessions, [Hershbein and Kahn \(2018\)](#) show that when demand for junior workers collapses, firms permanently elevate cognitive and educational skill thresholds. [Kahn \(2010\)](#) traces the supply-side fallout, demonstrating that cohorts entering the labor market during these contractions suffer decade-long wage losses and occupational downgrading. Together, these papers show that structural contractions in entry-level hiring impose durable human capital costs. [Deming and Noray \(2020\)](#) confirm these costs compound at the career level as job-specific skills depreciate among workers lacking foundational supervised experience. The AI mechanism replicates this exact human capital destruction with one critical difference. Where macroeconomic shocks suppress junior hiring *cyclically*, AI deployment provides the economic rationale to suppress it *permanently*.

3.3 Occupational Downgrading and Persistent Earnings Penalties

Individual-level occupational downgrading propagates beyond the point of initial displacement. [Gibbons and Waldman \(1999\)](#) formalize the career ladder as an assignment mechanism in which firms match workers to tiers based on revealed productivity signals, and promotion follows when output crosses the threshold for the next tier. Extending this logic to a market-wide shock yields a downward cascade prediction. When AI compresses the employment capacity of a given tier, workers above it do not absorb the loss in

place—they compete downward. The displaced senior worker takes a mid-level position; the incumbent mid-level worker takes an entry-level position; the entry-level worker can no longer find work. This structural crowding out leaves employment counts stable across intermediate tiers but compresses wages. As [Kambourov and Manovskii \(2009\)](#) demonstrate that human capital is strictly tied to a specific occupation rather than a firm or industry, meaning each step down this cascade actively destroys a worker’s accumulated skills.

The earnings consequences of this process are well-established and severe. [Jacobson et al. \(1993\)](#) show that workers displaced from high-tenure positions face earnings losses averaging 25% that show no sign of recovery. Crucially, these losses are concentrated among workers who subsequently find employment, indicating that the loss reflects a permanent drop in job quality rather than unemployment duration. [Davis and von Wachter \(2011\)](#) translate these deficits into lifetime present-value terms, demonstrating that displaced workers lose between 1.4 and 2.8 years of pre-displacement earnings depending on aggregate labor market conditions. However, the most directly relevant identification template is [Autor et al. \(2014\)](#). Exploiting cross-regional variation in trade exposure, they document individual-level earnings losses that fall overwhelmingly on workers in the lower half of the wage distribution. Because these losses occur through reduced within-year earnings rather than extended unemployment, they prove that structural adjustment costs are highly concentrated by skill level in ways aggregate statistics cannot capture. Finally, [Cortes \(2016\)](#) uses panel data to trace individual transitions, documenting that displaced middle-wage workers transition systematically to lower-wage occupations — confirming the cascade operates at the individual rather than purely compositional level.

Generative AI amplifies the severity of this cascade. Unlike the industrial automation studied by [Acemoglu and Restrepo \(2020\)](#), which targeted routine workers with limited downward amplitude, AI targets the top of the occupational hierarchy where the cascade possesses maximum reach. [Massenkoff and McCrory’s \(2026\)](#) finding that young workers in high-exposure fields face a 14 percentage point decline in job-finding rates captures only the terminal tier of this process. The ultimate consequence of this top-down shock is a permanent, market-wide devaluation of accumulated human capital, as mid-career professionals are forced to abandon their specialized skill stocks to compete for lower-tier roles.

3.4 The Boundary of the Firm and Contractual Fissuring

The three mechanisms established above share a common structural feature. They are invisible to unemployment statistics because displacement occurs *within* employment rather than at its boundary. Contractual fissuring provides the fourth mechanism manufacturing this invisibility, a process grounded in the fundamental economics of the firm. [Coase](#)

(1937) established that firms exist because the transaction costs of using the price mechanism make coordinating through employment cheaper than coordinating through open markets. This boundary condition shifts when coordination complexity falls. [Weil \(2014\)](#) documents the first wave of this shift. As digital tools reduced the cost of monitoring contractor performance, firms systematically converted salaried employment into outsourced and contract based arrangements. This shift eroded employment protections without registering in any unemployment statistic because the affected workers remained, in a formal sense, employed.

Generative AI initiates a second and more structurally complete wave of fissuring, because it does not merely reduce coordination costs — it makes task *boundaries legible*. When a knowledge worker’s role bundles heterogeneous tasks, that bundling solves a specific information problem. Because it is contractually intractable to specify performance on a fraction of an interdependent role, firms use long-term employment to secure the full bundle. As AI makes individual tasks modular and independently measurable, the information problem dissolves alongside the employment bundle itself. The economic prediction is direct — AI adoption in an industry should be followed by a rise in the share of non-W-2 arrangements without a corresponding rise in unemployment. [Katz and Krueger \(2019\)](#) already document a secular rise in alternative work arrangements concentrated disproportionately among the college-educated professionals [Massenkoff and McCrory \(2026\)](#) identify as most AI-exposed. This shows that the fissuring process is already underway in AI-exposed industries. Building on this proxy evidence, the Anthropic Economic Index ([Anthropic, 2025](#)) now provides the precise data infrastructure required to link this ongoing fissuring process directly to firm-level AI adoption.

The macroeconomic consequence of contractual fissuring is that it creates a structural blind spot in aggregate employment metrics. A former salaried analyst who becomes a contract analyst is employed and appears in no unemployment count, but has lost employer-sponsored health insurance, retirement contributions, and the implicit income insurance of a long-term employment relationship. [Goldschmidt and Schmieder \(2017\)](#), exploiting detailed German administrative records, show that domestic outsourcing produces wage reductions of 10–15% for workers performing the same tasks at the same location under a new employer. This penalty is driven entirely by exclusion from the firm-specific wage premia embedded in the original employment relationship. Because these structural income losses arrive without triggering the macroeconomic monitoring frameworks that govern policy response, they create a systemic vulnerability. Crucially, the macroeconomic demand trap formalized by [Hemenway Falk and Tsoukalas \(2026\)](#) is activated not by aggregate joblessness, but by exactly this type of structural income deflation.

4 Contribution to the Literature

This proposed research makes three contributions to the economics of technological displacement. *Theoretically*, it aims to formalize the downgrading cascade mechanism to show how the AI-driven displacement of high-skill workers generates a chain of credential-occupation mismatches propagating downward through the skill distribution. By demonstrating that this structural crowding out produces income losses sufficient to activate the demand trap formalized by [Hemenway Falk and Tsoukalas \(2026\)](#), the framework would reframe the reassuring early evidence of [Massenkoff and McCrory \(2026\)](#) as a predictable feature of the adjustment process rather than proof of labor market resilience. *Empirically*, the project proposes the first systematic test of AI-attributable occupational downgrading. The strategy will construct a credential-occupation mismatch measure linked to the Observed Exposure scores of [Massenkoff and McCrory \(2026\)](#) while exploiting cross-industry variation in AI adoption timing for causal identification, specifically designing this approach to close the measurement gap that prevents existing studies from detecting the primary adjustment channel in high-skill labor markets. *On policy*, I propose developing a set of leading policy indicators grounded in the established mechanisms. The planned metrics include entry-level hiring ratios disaggregated by credential intensity, mismatch indices built from existing CPS longitudinal data, and task portfolio breadth measures tracking within-occupation task compression. Because these indicators will rely on existing administrative and survey data, they aim to provide the practical institutional infrastructure required to detect structural labor erosion before the demand trap locks in.

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